

Budgeted Replanning in History-Dependent Games

Proof Appendix and Supplementary Materials

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Working status. Proof and implementation supplement; complete numerical tables and reproduction records will be integrated after the frozen experiment.

S1 Finite game and realization-plan formulation

Write a complete own path as $i = (i_0, \dots, i_{T-1})$ and the opponent path as $j = (j_0, \dots, j_{T-1})$. Both use a state-independent menu of size b . At an own update t , the information available for a new choice is exactly $(i_{<t}, j_{<t})$. All previously selected own commands have then executed; their contents are already in $i_{<t}$. The choice is an element of $\{0, \dots, b-1\}^{e-t}$, where e is the next own update or the horizon. The opponent's currently outstanding suffix is excluded. These information sets have perfect recall: every earlier own choice and every earlier observation is recoverable from the present prefix.

For each own information set I , let $\sigma(I)$ be its incoming own sequence and let $\sigma(I)a$ denote each block-extended sequence. Include an empty sequence with realization weight one. A realization plan satisfies

$$x_\emptyset = 1, \quad x \geq 0, \quad \sum_{a \in A(I)} x_{\sigma(I)a} = x_{\sigma(I)}.$$

Collect these equations as $Ex = e$; write the opponent equations as $Fy = f$. For every terminal pair (i, j) put its full trajectory payoff into the matrix entry indexed by its final own and opponent sequences. If more than one terminal node shares an index, add its chance-weighted contribution. There are no exogenous chance moves in the formation benchmark. The expected payoff is $x^\top Py$ and the value is

$$\max_{x \geq 0, Ex=e} \min_{y \geq 0, Fy=f} x^\top Py.$$

The inner problem has dual $\max_v f^\top v$ subject to $F^\top v \leq P^\top x$. Consequently the maximizer solves

$$\max_{x, v} f^\top v \quad \text{s.t.} \quad Ex = e, \quad x \geq 0, \quad F^\top v \leq P^\top x, \quad (1)$$

with unrestricted v . Finite game duality and perfect-recall realization equivalence justify this standard formulation [4]. Its size is linear in the represented game tree, which can itself grow exponentially in T .

For a numerical candidate (x, y) , nonnegative conditional child weights are normalized at every information set and multiplied by repaired parent reach. If proposed child mass is zero, use a uniform conditional distribution. This constructs feasible realization plans in exact arithmetic. Report the actual flow residual and the maximum repair magnitude. Two unrestricted responses then give

$$\min_{Fy'=f, y' \geq 0} x^\top Py' \leq V \leq \max_{Ex'=e, x' \geq 0} x'^\top Py.$$

Feasibility repair does not correct a wrongly specified information set. The independent tiny-game normal-form enumeration is therefore a separate test of model representation.

S2 Exact counterfactual pricing

Fix a causal opponent policy q . For each externally specified own path i , let $C_q(j \mid i)$ be the probability of opponent path j . Causality means that for any t and opponent prefix $j_{<t}$,

$$\sum_{j \geq t} C_q(j \mid i) \quad (2)$$

depends on $i_{<t}$ and not on any later own action. This follows by multiplying the opponent's conditional decision probabilities, or by averaging causal pure policies. An opponent may retain a private command suffix inside a block; the marginalization integrates over that suffix rather than revealing it.

Initialize $H_T(i, j) = A(i, j)C_q(j \mid i)$. If $t < e$ are successive own boundaries, recursively define

$$H_t(i_{<t}, j_{<t}) = \max_{a \in \{0, \dots, b-1\}^{e-t}} \sum_{z \in \{0, \dots, b-1\}^{e-t}} H_e(i_{<t}a, j_{<t}z). \quad (3)$$

The terminal entries are weighted by opponent realization probabilities, not by an own policy. Suppose all decisions from e onward have been optimized. At time t , the own block a must be fixed before z is observed. Thus z is summed for each candidate a ; the maximizing block is then selected. The probability of the already executed opponent prefix is independent of this candidate. It can remain as an unnormalized common factor throughout the recurrence. The induction hypothesis permits different optimal continuations at distinct newly observed prefixes at e . This proves (3) and the root best-response value. Any stored maximizing choice at a zero-weight prefix is admissible.

For $b \geq 2$, the number of weighted entries processed is at most

$$b^{2T} + b^{2(T-1)} + \dots + 1 \leq \frac{b^{2T}}{1 - b^{-2}}.$$

The payoff matrix and response arrays therefore take $O(b^{2T})$ arithmetic and memory. Constructing the physical payoff and repeating this procedure inside an equilibrium solver remain additional costs. This is a specialized history-indexed response calculation, not a polynomial algorithm in the compact physical-state description.

S3 Restricted policy generation and stopping

Against a committed opponent, a feasible pure own policy p_m induces one own path $i_m(j)$ for each complete opponent path j . Form a restricted matrix with entries $B_{mj} = A(i_m(j), j)$, retaining every opponent column. Let w and q be restricted equilibrium distributions. The mixed own policy has full-game lower guarantee

$$L = \min_j \sum_m w_m B_{mj},$$

and exact pricing against q gives $U = \text{BR}_S(q) \geq V(S, C)$. Hence $[L, U]$ is valid. If $U = L$ in exact arithmetic, the restricted mixture and q are an equilibrium of the complete game. If $U > L$, a maximizing response cannot be a previously available row, since all old rows have payoff at most the restricted value against q . Adding that response strictly enlarges the finite set of available policies. The exact procedure terminates after finitely many additions, but this worst-case count can be exponential. This is the standard response-oracle argument [1, 3].

The implementation attempts at most 64 policy additions before switching to the complete realization-plan LP for the same game. The first attempt's time is included. A duplicate response with a numerical gap is reported as a stall, not used as a proof of convergence. All positive mixture weights enter full-response evaluation; pruning tiny positive weights would require an additional error budget.

S4 Coarsening a full-update behavioral strategy

Let $\pi_u(a \mid i_{<u}, j_{<u})$ be a full-update source strategy, completed by uniform distributions at its zero-reach histories. Fix one legal reference opponent action $r_\star$, the straight action in the experiment. For a target block $[t, e)$ and actual observed prefixes $(i_{<t}, j_{<t})$, define

$$\tilde{\pi}_{t:e}(a_t, \dots, a_{e-1} \mid i_{<t}, j_{<t}) = \prod_{u=t}^{e-1} \pi_u(a_u \mid i_{<t} a_{t:u}, j_{<t} r_\star^{u-t}). \quad (4)$$

For each fixed earlier own block prefix, the last factor sums to one over a_{e-1} . Repeating this argument backward proves normalization of (4). Its factors use only observed prefixes, private sampled own commands and a deterministic hypothetical continuation. No actual future opponent command is supplied. State-independent menus make all resulting blocks legal. Thus the construction is a feasible target-calendar policy.

The source need not be optimal for this feasibility result. If it is obtained as a root mixture, first convert its realization distribution to behavioral kernels using perfect recall. Deleting updates after this conversion defines a new behavioral strategy; it need not preserve the source's mixture over latent pure plans or its outcome distribution. This distinction matters at histories that become newly reachable after coarsening.

Let U_F be any valid upper bound on the full-update value for the same opponent class. A full response to $\tilde{\pi}$ gives L' . Policy inclusion and feasibility imply

$$0 \leq V(F, R) - V(S', R) \leq U_F - L'.$$

To evaluate the minimizing response, apply (3) to the inverted matrix $-A^\top$ weighted by the counterfactual realization law of $\tilde{\pi}$, using the opponent calendar R . This directly quantifies the actual projected policy. It avoids unproved smoothness assumptions but can yield a loose bound. It does not strengthen the general abstraction decomposition of Kroer and Sandholm [2].

S5 Calendar search and transferable opponents

A search node consists of included epochs I and undecided epochs D . Excluded epochs are the complement of $I \cup D$. Its remaining family is

$$\mathcal{C}(I, D) = \{S : I \subseteq S \subseteq I \cup D, |S| \leq K\}.$$

For any S in this family, $V(S, R) \leq V(I \cup D, R)$ by policy inclusion. The latter game is a valid relaxation even if $|I \cup D| > K$. Its numerical upper bound is therefore a node upper bound. A fixed opponent strategy remains feasible at every node because its own calendar, available history and action menu are unchanged. Thus $\text{BR}_{I \cup D}(q)$ is another valid upper bound, and a minimum over such bounds is safe.

If an upper bound is at most the incumbent feasible lower bound plus ε , prune the node while retaining that bound in the terminal bound list. Otherwise split on one undecided epoch, producing the included and excluded child. These children partition the remaining family. When $|I| = K$, only I remains. When $|I \cup D| \leq K$, inclusion implies that $I \cup D$ dominates all calendars within the node, so solve only that calendar. After finite exhaustion, every feasible calendar has either been evaluated or belongs to a represented pruned node. The largest retained upper bound, together with the best feasible lower bound, encloses W_K . If unresolved inner solves leave a gap above ε , the search result is unresolved. Finite termination alone does not establish numerical optimality.

S6 A sufficient modular special case

Suppose there are independent component games indexed by potential update epochs m . Component m has maximizing pure-policy set P_m^0 without its update and $P_m^1 \supseteq P_m^0$ with its update, a fixed minimizing set Q_m , and payoff $a_m(p_m, q_m)$. Assume the whole game's pure-policy sets are exactly the Cartesian products of the component sets and its payoff is the sum of the component payoffs. These are structural assumptions: there are no shared opponent resources, inter-component physical state constraints or additional cross-component observations that enlarge these sets.

Let v_m^d be the minimax value of component m under $d \in \{0, 1\}$. The product of component maximizing equilibrium policies guarantees $\sum_m v_m^{\mathbf{1}\{m \in S\}}$ against every joint minimizing policy, because each of its component marginals is feasible. The product of component minimizing equilibria gives the matching upper bound against every joint maximizing policy. Therefore

$$V(S) = \sum_m v_m^0 + \sum_{m \in S} (v_m^1 - v_m^0),$$

which is modular. The assumptions define a checkable special class and are not claimed for the formation benchmark. The shared trajectory and target history prevent this product factorization in general.

S7 Boundary conditions and historical counterexamples

The angular kernel changes firing sector at specified bearings. A nonzero one-sided jump is incompatible with a finite global Lipschitz constant: the ratio of the jump to an arbitrarily small bearing separation is unbounded. All current certificates operate on the explicitly evaluated finite payoff matrix, including its boundary convention. They do not certify angular-grid refinement, continuous-control limits or the physical accuracy of the kernel. The archived floating-boundary amendment groups values within 10^{-8} degrees at the designated ± 30 and ± 150 degree boundaries. No payment matrix is projected onto an antisymmetric matrix to pass a mirror test.

Exchange oddness is not a Bellman collapse. Take absorbing states $s = \pm 1$ with no actions and stage reward s . Exchanging players sends s to $-s$, but the two-step value is $2s$, not the one-stage reward. A separate issue is pure versus mixed play: the rock–paper–scissors antisymmetric matrix has pure max–min value -1 and mixed minimax value zero. Zero-value claims must identify the correct exchange-fixed state and the matching policy classes.

Changing a forced operating speed changes the feasible trajectories and is not a nested enlargement of the action menu. Even an actually nested ability set guarantees only that each value with fixed opposing class cannot decrease; it does not make the difference of two such values monotone. Archived v12 range and v13 mobility contrasts therefore remain separate negative evidence, not premises of the present budget theorem.

S8 Numerical intervals and decision logic

The scientific objective is the finite deterministic game. Declared numerical intervals come from feasible strategies, complete responses, local error propagation and explicit floating-point padding. They are not interval-arithmetic proofs: arithmetic and physical-payoff rounding are not enclosed with directed rounding. The implementation reports this scope with every solver result.

For adaptation retention ρ , define $M_K = W_K - \rho V_F - (1 - \rho)V_0$. Given value intervals, the lower and upper margins are

$$M_K^- = L_K - \rho U_F - (1 - \rho)U_0, \quad M_K^+ = U_K - \rho L_F - (1 - \rho)L_0.$$

A nonnegative lower margin proves sufficiency, and a negative upper margin proves insufficiency. The smallest possibly sufficient budget and the smallest certified sufficient budget are reported separately. Equality of these budgets certifies minimality. When the adaptation interval is too close to zero, the reported quantity is absolute loss; no unstable retention percentage is formed. These are optimization intervals on a deterministic finite design, not sampling confidence intervals.

References

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